

Counting to Infinity: An Introduction to the “Absolute Numeral System”

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Abstract. ‘Infinity’ has been one of the most challenging as well as the most widely important term for a lot of mathematicians, cosmologists or the scientists dealing with micro or macro levels in the field of technological advancements or research. The term “infinity” is being extensively used in numerous mathematical explanations, assumptions, calculus, modern physics, quantum physics, astronomy, cosmology, set theory, modern number systems etc. In spite of being such a popular concept used in modern mathematics, it is still ‘undefined’. Yet, the urge to find the actual infinity is still high by seeing the requirement of the today’s advancement in quantum physics, particle physics etc. Taking references from the ancient *Jaina* scriptures and literature initially written around in 200 B.C. and subsequently developed till today; the ‘Infinity’, which was coined centuries back, has been re-researched and is effectively translated into today’s mathematical notations. A new concept of numeral system has been introduced through this paper called “Absolute Numeral System”, for dealing with the large value ranges that includes Infinity. The system defines and differentiates the numbers into 3 major classes; numerable, Innumerable (and its major 3 types) and Infinite (and its 3 major types) in great detail. ‘Four Cylinder Hypothetical Visualization’ method, a new function ‘*Shalakatrāyṇishṭapana*’, various other algorithms, pseudo-codes are used to explain the Innumerable and Infinite Class for in-depth understanding. Another thing to note that numbers that are obtained from the Innumerable and Infinite class are essentially big enough to surpass the “Graham’s Number” (the largest number discovered yet in the Universe). Also, the idea of ‘Finite Infinity’ is provided with an example of Matter-Transmigration. This paper effectively guides through the method to ‘count’ till infinity by using various new mathematical concepts, notations, functions, methods and cosmological inferences.

Keywords: Absolute, Cosmology, Infinity, Innumerable, Jainism

Introduction

The term “infinity” is being extensively used in numerous mathematical explanations, assumptions, calculus, modern physics, quantum physics, astronomy, cosmology, set theory, modern number systems etc. Without the term “infinity”, it is almost impossible to draw out any valuable conclusions from any of fields that includes use of infinity in the calculations. Though, infinity being a very important aspect in modern world science and mathematics; it is relative, ill-defined and the concept of infinity has just bounded itself to the fact that “Infinity means something larger than a given n ” or “There is no actual-infinity, it is just a collection to which we can add unceasingly”[1] There is an idea that deals with “Potential Infinity” and “Actual Infinity” as in [2] where Actual Infinity is the idea developed to at the point where our intelligence fails; while Potential Infinity being a logic that even it is infinite, we know it is potentially existing logically like the fact that there is another natural number after infinite-th natural number; while also there has been discussions about the concepts of limits, straight lines, etc. to define the term ‘Infinity’. There are various Ancient philosophies including Greek, Indian etc. as in [3] where Aristotelian theory suppresses the concept of “actual” infinity. While, Cantor has defined infinity to a certain extent as in [4] and has well-defined the concept of ‘infinite sets’, but still the concept of Infinity, yet is undiscovered and undefined. The infinity is in itself has become a relative term and is depending on frame of reference.

History of the “Absolute Numeral System”:

The history of Jainism’s scriptural and literary works has been notable and has contributed to Indian Literature. Taking intimations from the metaphysics and cosmos that is being referenced from the *Jaina* Literature as described in [5]; this paper guides through the ancient number system, which is redefined as “Absolute Numeral System”; available in written form for the first time in 200 B.C. in order to explain in detail many of the cosmological, astronomical, philosophical and metaphysical aspects of Science; from which some of them will be given as examples as an application of this Numeral System. One of the oldest origins found of this “Absolute Numeral System” is in the scripture “*Shatkhandāgama*” written by *A.Bhutbali* and *A. Pushpadanta* in around 200 B.C.; written on palmscripts, is today available in original Prakrit language along with Hindi translations. Major contributions includes *A.Umaswami*’s “*Tatvarthasutra*” which had similar explanations and references to the Numeral

System in the 1st Century A.D., *A. Nemichandra* in 9th Century wrote various mathematical as well as cosmological literary works like “*Triloksara*”, “*Gommatsara-JeevKaand*”, “*Gommatsara-Karmakaand*”, “*Ganitsaara*” etc. where many mathematical concepts[6] like logarithm(named as ‘*ardhachheda*’ in *Shatkhandagam*), permutations and combinations, probability, extended propositional logic, infinite series and solutions etc. were introduced.

1. The Absolute Numeral System

An Absolute numeral system is a writing system for expressing numerical quantities; that is, a mathematical notation for representing numbers of a given set; specifically used for representation and defining the scale for large and incomputable numbers which are ‘relatively’ considered as Infinity in the present number system. The Absolute Numeral System is divided broadly into three class, (1) Numerable(C) (2) Innumerable(I) (3) Infinite(A)

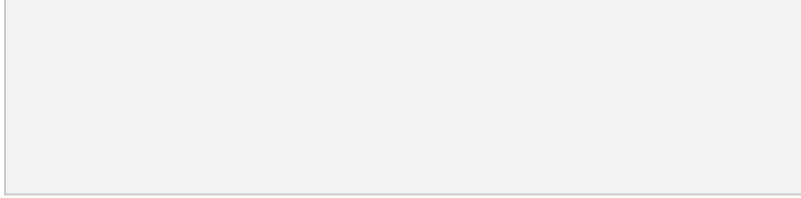


Fig. 1. Classification of the Absolute Numeral System

1.1 Some Definitions and Notations:

Absolute Universal Class: The class containing all the elements starting from 2 to infinity. It is represented as U.

$$U = \{\min(C), \dots, \max(C)\}, \{\min(I(1)), \dots, \max(I(3))\}, \{\min(A(1)), \dots, \max(A(3))\}$$

Where,

$$C = \{n \mid n \in \mathbb{N}, \min(C) \leq n \leq \max(C)\}, \text{ where } C \text{ represents Numerable Class}$$

$$I = \{n \mid n \in \mathbb{N}, \min(I(1)) \leq n \leq \max(I(3))\}, \text{ where } I \text{ represents Innumerable Class}$$

$$A = \{n \mid n \in \mathbb{N}, \min(A(1)) \leq n \leq \max(A(3))\}, \text{ where } A \text{ represents Infinite Class}$$

Class: A class is a collection of well-defined sets which fall into a particular group of elements having similar properties. All the classes of the Absolute Numeral System are subclasses of the Absolute Universal Class.

Class-type: The subclass of the Class is called Class-type. It is represented as Class(type). For example, Class ‘A’ has type ‘1’, it is represented as A(1).

Minima of a Class-type: The smallest element corresponding to a respective Class-type is called Minima(*jaghanya*) of a Class-type. It is represented by $\text{Class(type)}_{\min}$ or $\min(\text{Class(type)})$.

Maxima of a Class-type: The largest element corresponding to a respective Class-type is called Maxima(*utkrushtha*) of a Class-type. It is represented by $\text{Class(type)}_{\max}$ or $\max(\text{Class(type)})$.

Intermediate of a Class-Type: The subset of the respective class type that includes all the elements in the Class-type excluding the minima and maxima elements is called Intermediate(*madhyama*) of a Class-type. It is denoted as $\text{Inter}(\text{Class(type)})$.

$$\text{Inter}(\text{Class(type)}) = \{n \mid n \in \mathbb{N}, \min(\text{Class(type)}) < n < \max(\text{Class(type)})\}$$

2. Numerable Class

Numerable Class(*samkhyaat*): The set of all the natural numbers starting from 2 to

$(\text{Innumerable}(\text{Type}-1))_{\min} - 1$ fall into this class. It doesn’t have any Class-type. In absolute numeral system, it can be represented in set roster-form as:

$$C = \{2, 3, 4, 5, \dots, (\text{Innumerable}(\text{Type}-1))_{\min} - 1\}$$

It can be represented in set-builder form as:

$$C = \{n \mid n \in \mathbb{N}, \min(C) \leq n \leq \max(C)\}$$

Where,

$$(\text{Numerable})_{\min} = 2 \text{ or } \min(C) = 2$$

$$(\text{Numerable})_{\max} = (\text{Innumerable}(\text{Type}-1))_{\min} - 1 \text{ or } \max(C) = \min(I(1)) - 1$$

The Numerable Class lists the numbers which can be perceived or understood just by representing with digits or symbols. A threshold is being set taking into consideration by the absolute parameters of Cosmology of Universe. For example, if one is asked to tell difference between 10 and 1000, it can be understood and felt by an individual. But the difference between 10^{100} and 10^{1000} can’t be understood and is considered as “Relative Infinity” with respect to present frame of reference; while here absolute threshold is made. For example, the number of humans in the

entire Universe according to Jainism cosmology belongs to numerable class which is accurately 79,228,162,514,264,337,593,543,950,336; the number of stars in the Universe are $2 \times 68975 \times 10^{14}$; the life span of Lord Rishabhdeva was $84 \times 10^5 \times 7056 \times 10^{10}$ years (can be seen as average life span of people in the starting of 4th Cosmic age), etc. are some example of huge numerical quantities that are ‘numerable’ and listable.

3. Innumerable Class

Innumerable Class(*asamkhyaat*): The group of well-defined set of elements which are unlistable or innumerable but not infinite; and which cannot be understood if represented in form of digits or symbols, but by increases or decreases by performing arithmetic operations like addition, subtraction etc. on it and is not endless. By using Hypothetical visualization techniques, assumptions, situational explanations and methodologies the large innumerable values of this class can be explained. The class-types of the Innumerable Class can be effectively defined by using “Four-Cylinder Hypothetical Visualization” method.

The Innumerable class (denoted as I) can be divided into 3 Class-types as:

Type-1(*paritasamkhyaat*), Type-2 (*yuktasamkhyaat*) and Type-3(*asamkhyaat**asamkhyaat*). Each class-type has its respective minima, maxima and intermediate subsets.

In absolute numeral system, the Innumerable Class can be represented in set roster-form as:

$$I = \{ \{ \max(C)+1, \dots, \max(I(1)) \}, \{ \max(I(1))+1, \dots, \max(I(2)) \}, \{ \max(I(2)) + 1, \dots, \max(I(3)) \} \}$$

Or

$$I = \{ \{ \min(I(1)), \dots, \min(I(2)) - 1 \}, \{ \min(I(2)), \dots, \min(I(3)) - 1 \}, \{ \min(I(3)), \dots, \min(I(4)) - 1 \} \}$$

It can be represented in set-builder form as form of individual class-types as:

$$I(1) = \{ n \mid n \in \mathbb{N}, \min(I(1)) \leq n \leq \max(I(1)) \}$$

$$I(2) = \{ n \mid n \in \mathbb{N}, \min(I(2)) \leq n \leq \max(I(2)) \}$$

$$I(3) = \{ n \mid n \in \mathbb{N}, \min(I(3)) \leq n \leq \max(I(3)) \}$$

3.1 Four-Cylinder Hypothetical Visualization:

The Four-Cylinder Hypothetical Visualization method[7] is used to define the elements of Innumerable Class. This Visualization technique of explanation of Innumerable includes 3 major components.

- Four Cylinders (initially empty) namely A(of variable diameter)(*anyavastha*), B(*shalaka*), C(*pratishalaka*) and D(*mahashalaka*)
Initial diameter of each cylinder: 4×10^8 miles (10^5 *yojanas*)
1 *yojana* = 4000 miles = 6437376 m
Height of the cylinder: 4×10^6 miles (10^3 *yojanas*)
- Spherical seeds of diameter: 0.29mm (the volume of the seeds is not much important as this is a hypothetical imagination-based method. The seeds are often considered to be of ‘mustard’(*sarso*) to ease the understanding of its size)
- Relatively Infinite plane consisting of Concentric Circles (actually the depiction of the ‘Middle Realm’ of the Universe) with the radius getting doubled for each circle from the center.

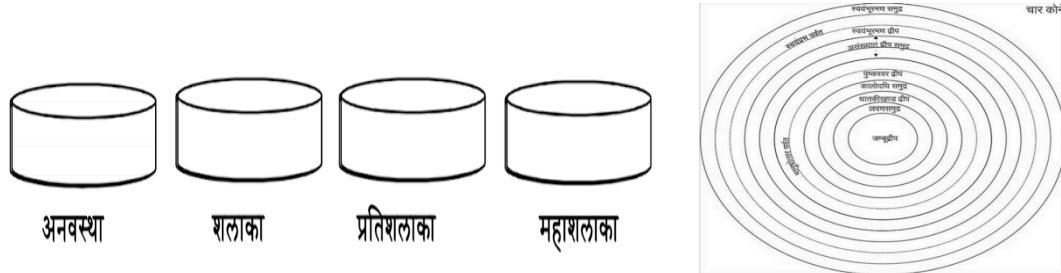


Fig. 2. The above figure explains the two major parts of ‘Four Cylinder Hypothetical Visualization Method’: first the hypothetical cylinders that are to be made (along with their ‘sanskrit’ names) and adjacent to it is the infinite plane of concentric circles.

Some important functions and terms that needs to be considered while understanding the algorithm:

$$\min(I(3)) = \min(I(2))^2 \quad (1)$$

3.2 Some different types of Innumerable

Innumerable(U): To compute the value of Innumerable(U) (*lok-praman asamkhyaat*), few things are needed to be considered and assumed. The Universe is considered to be finite and not infinite and occupies a definite volume of the total spatial region. As the indivisible part of matter is considered as ‘atom’, the indivisible (not hypothetically infinitesimal) part of the space is defined as ‘space-element’(*pradesh*). As the total space occupied by the Universe is finite, the total number of space-elements of the Universe cannot be ‘infinite’ and hence fall in the ‘Innumerable’ class. Hence, Innumerable(U) is that total number of space-elements of the Universe.

U(Innumerable(U)): The value of U(Innumerable(U)) (*asamkhyaat lok-praman*) is computed by considering ‘Innumerable’ number of hypothetical Universes and then adding all the space-elements of each of the respective Universe.

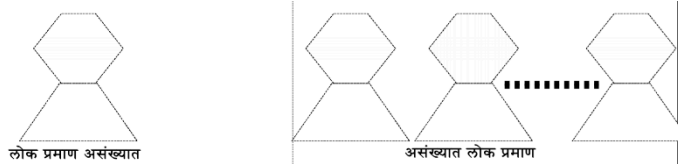


Fig. 3. The figure explains the difference between the two different Innumerable Types; (1) Innumerable(U) and (2) U(Innumerable(U))

Some other types of Innumerable are *palya*, *saagar*, *sachyangul*, *pratarangul*, *ghanangul*, *jagatshreni*, *jagatpratar*, *jagatghan* etc. as described in [9].

4. Infinite Class

Infinite Class(*anant*): The group of well-defined set of elements where all the members of the sets are ‘Infinite’ in nature. Infinite refers to the values that are endless and remain unaffected by the carrying out the mathematical operations like addition, subtraction, multiplication, division etc. by the members of numerable or innumerable class.

In other words, the numbers in this class don’t decrease on subtraction or increase with addition; they are “relatively” unbounded which literally means that the end points of the quantities that has a value belonging to “infinite” class doesn’t exist.

For example; if the “age” of the Universe is considered infinite, then it can be understood that there is no starting or ending point of the Universe and hence; the Universe is self-existing and uncreated since infinite time and will never collapse up till infinite time (i.e. never, as it is unbounded). But even though the Infinity is not ill-defined. Similar to the “Innumerable Class”, there are a set of methods and algorithms to define each set of Infinite Class.

In absolute numeral system, the Infinite Class is divided into 3 types;

Type-1(*paritanant*), Type-2(*yuktanant*) and Type-3(*anantanant*) can be represented in set roster-form as:

$$A = \{ \{ \max(I(3))+1, \dots, \max(A(1)) \}, \{ \max(A(1))+1, \dots, \max(A(2)) \}, \{ \max(A(2)) + 1, \dots, \max(A(3)) \} \}$$

Or

$$A = \{ \{ \min(A(1)), \dots, \min(A(2)) - 1 \}, \{ \min(A(2)), \dots, \min(A(3)) - 1 \}, \{ \min(A(3)), \dots, \max(A(3)) \} \}$$

Each Class-type can be represented in set-builder form as:

$$A(1) = \{ n \mid n \in \mathbb{N}, \min(A(1)) \leq n \leq \max(A(1)) \}$$

$$A(2) = \{ n \mid n \in \mathbb{N}, \min(A(2)) \leq n \leq \max(A(2)) \}$$

$$A(3) = \{ n \mid n \in \mathbb{N}, \min(A(3)) \leq n \leq \max(A(3)) \}$$

Before defining the infinity, first let us define an important function that will be used to “Count till Infinity”. The name of the function is “*shalakatranyishtapana*” as in [10]. It can be abbreviated as ‘STN’ for our ease. Tabular method is used to effectively describe the function.

4.1 Shalakatranyishtapana(STN)

The function ‘STN’ takes three passes. In the each of the three pass, the input number $x \in \mathbb{N}$ is decremented upto zero each time by one and subsequently calculating the value at each x^{th} count as $:(x-1)^{(x-1)}$. The output is the value

when count is zero; and that value is the input value for the next pass. For the first pass, the input value x is the number whose STN is to be computed.

Let us calculate the *shalakatranyishtapana* of the number '2' by using tabular method:

Table 2. *shalakatranyishtapana* pass 1/3

Count(Shalaka)	Base	Index	Value(Base ^{index})
2	2	2	4
1	4	4	256
0			

Table 3. *shalakatranyishtapana* pass 2/3

Count(Shalaka)	Base	Index	Value(Base ^{index})
			256 ²⁵⁶
256	256	256	(3.3231700E616)
255	256 ²⁵⁶	256 ²⁵⁶	256 ²⁵⁶ ^ 256 ²⁵⁶
254	256 ²⁵⁶ ^ 256 ²⁵⁶	256 ²⁵⁶ ^ 256 ²⁵⁶	(256 ²⁵⁶ ^ 256 ²⁵⁶) ^ (256 ²⁵⁶ ^ 256 ²⁵⁶)
253	.	.	.
.	.	.	.
.	.	.	.
.	.	.	=X
0			

Table 4. *shalakatranyishtapana* pass 3/3

Count(Shalaka)	Base	Index	Value(Base ^{index})
X	X	X	X^X
X-1	(X^X) ^ (X^X)	(X^X)	(X^X) ^ (X^X)
X-2	(X^X) ^ (X^X)	(X^X) ^ (X^X)	((X^X) ^ (X^X)) ^ ((X^X) ^ (X^X))
X-3	.	.	.
.	.	.	.
.	.	.	.
.	.	.	=Y
0			

$$STN(2) = Y \quad (1)$$

It can be understood from the above tables, that the value "Y" is consisting of innumerable digits.

Now, imagine about the value obtained by performing STN of the number 3, of the number 4, of 10¹⁴ and so on. In comparison to Graham's number (currently claimed the largest number of the Universe) [11]. The numbers that are obtained by performing "*shalakatranyishtapana*" are so big, which can possibly surpass the Graham's number. Also, this function can be claimed as the largest value outputting function. Still, this is not infinity; because as discussed Infinity is larger than we consider it to be. The algorithms [12] to calculate the infinite quantities is discussed now.

4.2 Algorithm to calculate min(A(1)) i.e. minimum value of Infinity type-1:

Step 1: Perform *shalakatranyishtapana*(max(I(3))). The value obtained is A(say).

Step 2: Add the following values to A

(i) $4 * \text{Innumerable}(U)$

(ii) $2 * \text{Innumerable}(U(\text{Innumerable}(U)))$

The number obtained by addition of the six quantities to A= B(say)

Step 3: Perform *shalakatravnishthapana*(B). The value obtained is C(say)

Step 4: Add the following values to C

(i) $20 \times 10^{14} \times \text{Innumerable}(saagar)$

(ii) $3 \times \text{Innumerable}(U(\text{Innumerable}(U)))$

The number obtained by the addition of the four quantities to C is D(say)

Step 5: Perform *shalakatravnishthapana*(D). The value obtained is E.

Thus,

$$\min(A(1)) = E \quad (1)$$

The minimum value of the set of Infinity type-2;

$$\min(A(2)) = \min(A(1))^{\min(A(1))} \quad (1)$$

The minimum value of the set of Infinity type-3;

$$\min(A(3)) = \min(A(2))^2 \quad (1)$$

4.3 Algorithm to calculate $\max(A(3))$ i.e. maximum value of Infinity type-3(the last value of the Absolute Numeral System):

Step 1: Perform *shalakatravnishthapana*($\min(A(3))$). The value obtained is A(say).

Step 2: Add the following values to A

(i) Total number of Gods (the souls who have achieved salvation, i.e. supreme omniscient beings(*siddhas*)) (=Infinite)

(ii) Total number of souls who are in *nigoda* life-form (excluding the vegetation-nigodas life-form)

(iii) Total number of souls who are in *nigoda* life-form (including the vegetation-nigodas life-form)

(iv) Total number of ‘ultimate-particles’ (*parmanu*’ as per Jainism) of all existing matter in the Universe

(v) Total number of time states of the Universe (on representing the Universe on a time-line; each dt represents a unit time state; and as discussed age of the Universe is “infinite” So adding all the time states, we get the total time state)

(vi) Total number of space-elements(*pradesh*) of the Space(*akash*) (including of both; Universe(*loka*) and Non-Universe(*aloka*))

Each value of the above listed ‘six’ quantities is infinite times the previous one. That is, the number of souls in *nigoda-life-form* is infinite times than the number of Gods; and the total number of space elements of the Space (Including Universe and Non-Universe) is infinite times the number of time states of the Universe.

The number obtained by adding the above infinite quantities is B(say).

Step 3: Perform *shalakatravnishthapana*(B). The value obtained is C(say)

Step 4: Add the ultimate elements of the *dharma* and *adharma* (the ‘motion’ and ‘rest’ existents) to C. The value obtained is D

Step 5: Perform *shalakatravnishthapana*(D). The value obtained is E.

Step 6: $F = K - E$

Where K is $\max(A(3))$

Step 7: $F + E = K = \max(A(3))$

This step signifies that even though performing a lot of operations and calculations, the $\max(A(3))$ is still so big that the obtained value ‘E’ is an infinitesimal quantity with respect to the maximum of Infinity type 3; the last number of the Absolute Numeral System. For example; one wants to count up to 1000 and has managed to count up to 100 and so subtract 100 from 1000 and add 900 into 100 to get 1000 which symbolizes is that in spite of these large computations the number E is far less than $\max(A(3))$.

The question arises what quantity of the Universe has the $\max(A(3))$. To explain, understanding the concept of Karma Theory [13] is required. In brief, there are a type of karmic particles attached to every soul which puts a knowledge barrier on the soul’s power to perceive things. A person (all living entities are considered as souls) having karmas related to Knowledge-barrier perceives an object X; lets define the perceiving intensity as I. This intensity is different for person to person depending on individual’s karmic particle bondage at a particular time ‘dt’. So, for someone who is free from the karmic bondage has zero knowledge barrier and subsequently the intensity I

will be Infinity. So, the Omniscient being has the $\max(A(3))$ intensity of Knowledge and hence in each 'dt' time interval the being can know the entire Universe simultaneously.

5. Concept of Finite Infinity and Transmigration:

The Infinity is defined as unbounded and endless; and hence the concept of "Finite Infinity" seems to be contradictory with the definition. Finite Infinity literally means that though it being Infinite, it is being completed at some point. There is a theory of Transmigration [14]; which are of 5 types, in which one of them is discussed in a very superficial method. Let at a particular time 'dt', a soul by its karmic flux gains some matter particles(atoms) with the particular intensity units of touch, taste, smell and color. Take the first six atoms (the ultimate particle as per Jainism) in order that are being gained by the soul as a part of physical body or materials. After that 'dt' time, it will lose all the six particles and gain another with other intensity units. Each and every atom in the universe is considered to be distinguishable particle; and hence the six out of Infinitely many total matter particles in the Universe (Probability: $6/\infty = 0$). The time required to gain the same six particles in same order with exactly same intensity units is called 1 Matter-Transmigration. It is believed that the soul has done 'Infinite' Matter-Transmigrations till now; which means though it takes infinite time for completing that; the time though being infinite completes at a certain point to achieve same combination of 6 particle, and it hence becomes relatively finite as it cannot be regarded as endless now.

Conclusion

From this paper, the detailed description of the "Absolute Numeral System" has been provided to define the actual 'Infinity', and not making an abstract concept anymore. Various algorithms, visual and pseudo-code techniques, tabular methods, functions and equations are used to explain possibly very large numbers which are on the number line; even larger than Graham's number. By taking in consideration about the presented concept of Infinity; it can be used in numerous applications in quantum physics, astronomy, cosmology, mathematics, technological advancements and other scientific foundations [15]. The Infinity is a wide concept and definitely still a field for in-depth research and apply it to the modern mathematical solutions.

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